

CS5351 2025/26 学期A

现代代码审查

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课程大纲

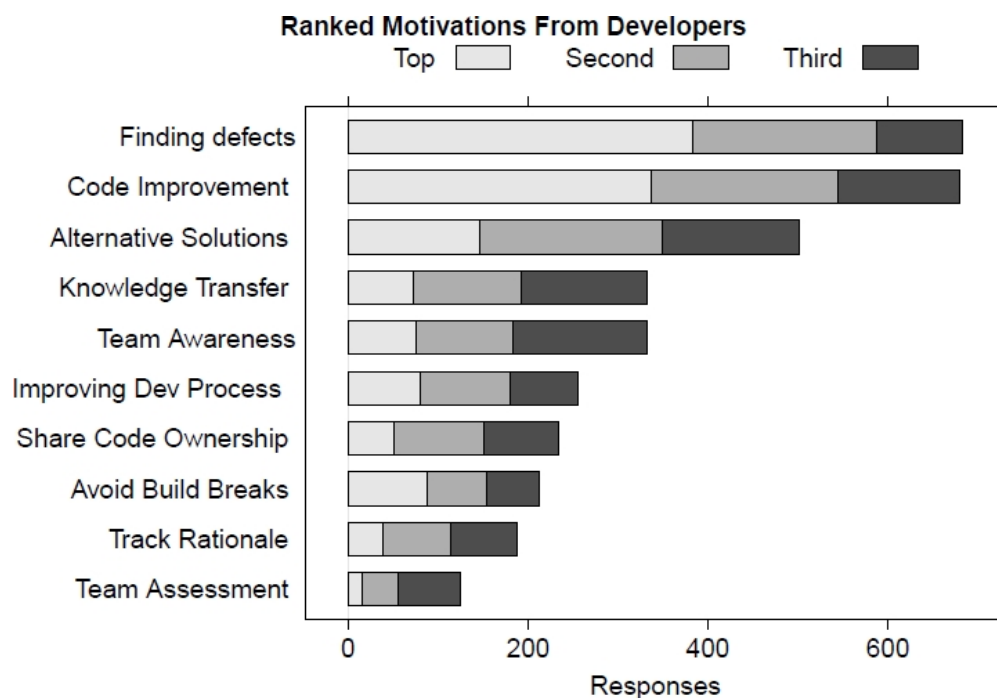
- ◆ 什么是代码审查?
- ◆ 为何需要代码审查?
- ◆ 代码检查
- ◆ 现代代码审查
- ◆ 背景阅读
 - https://en.wikipedia.org/wiki/Fagan_inspection
 - <https://ieeexplore.ieee.org/document/6606617/>
 - 连接校园网络时可下载
- ◆ 尝试学生团队开发的代码审查工具（如MCR-2）

什么是代码审查

- ◆ 由非代码作者的开发者对源代码进行人工审查。
- ◆ 是软件工程领域成熟的实践方法
- ◆ 旨在通过知识转移来**维持和提升源代码质量**，并通过知识转移来**维持开发社区**
他人采用的设计与实现方案[12]

什么是代码审查

◆ 由非代码作者的开发人员对源代码进行人工检查。



除发现缺陷外，代码审查还具有多重目的：代码优化、替代方案探索、知识传承、团队建设等。

Fig. 3. Developers' motivations for code review. [1]

不同层级的正式性

◆ 代码演示

- 非正式会议，程序员带领审查团队审阅其代码，审查者尝试找出缺陷
- 若程序员与评审者非同一人，效果更佳
- 有时可与结对编程结合

◆ 代码检查

- 正式会议
- 能有效发现代码中的错误。
- 课后阅读：https://en.wikipedia.org/wiki/Fagan_inspection

◆ 现代代码审查

```

bool isWin(char game[3][3]){
    bool win = false;
    if (game[0][0] == game[0][1] && game[0][1] == game[0][2] && (game[0][0] == 'X' || game[0][0] == 'O')) win = true;
    if (game[1][0] == game[1][1] && game[1][1] == game[1][2] && (game[1][0] == 'X' || game[1][0] == 'O')) win = true;
    if (game[2][0] == game[2][1] && game[2][1] == game[2][2] && (game[2][0] == 'X' || game[2][0] == 'O')) win = true;
    // 纵列
    if (game[0][0] == game[1][0] && game[1][0] == game[2][0] && (game[0][0] == 'X' || game[0][0] == 'O')) 获胜 = true;
    if (game[0][1] == game[1][1] && game[1][1] == game[2][1] && (game[0][1] == 'X' || game[0][1] == 'O')) win = true;
    if (game[0][2] == game[1][2] && game[1][2] == game[2][2] && (game[0][2] == 'X' || game[0][2] == 'O')) win = true;
    // 对角线
    if (game[0][0] == game[1][1] && game[1][1] == game[2][2] && (game[0][0] == 'X' || game[0][0] == 'O')) 获胜 = true;
    if (game[0][2] == game[1][1] && game[1][1] == game[2][0] && (game[0][2] == 'X' || game[0][2] == 'O')) win = true; return win;
}

int main(){
    int i, j;
    char game[3][3] = {' '}; // 井字棋char player1 = 'X';
    char player2 = 'O';
    bool turn = true; // 玩家1回合时为false, 玩家2回合时为true. 玩家1先手. cout << "X = 玩家1" << endl << "O = 玩家2" << endl;
    for (int n=0; n<9; n++){
        turn = !turn; // 使用非运算符将true转为false或false转为true. if (turn == false)

        cout << "玩家1: " << endl;

        else

        cout << "玩家2: ";

        cout << "选择标记哪个单元格? i:[1..3], j:[1..3]: "; cin >> i >> j;
        if (turn == false) game[i][j] = 'X';
        else
            game[i][j] = 'O';
        if (isWin(game)){
            cout << "获胜! " << endl;
            break; // 需要终止问题
        }
    }

    if (i==3) // 所有i=0到2的单元格已输入但未获胜 cout << "平局! " << endl;

    // 将游戏状态输出至控制台
    cout << game[0][0] << " " << game[0][1] << " " << game[0][2] << endl;
    cout << game[1][0] << " " << game[1][1] << " " << game[1][2] << endl;
    cout << game[2][0] << " " << game[2][1] << " " << game[2][2] << endl; return 0;
}

```

代码解析：

isWin() 的目标是实现什么功能，因此.....

Main() 在这几行代码中执行此操作，在那些行中执行那项操作

代码解析

- ◆ 来自一篇旧论文的结果[21]
 - 类似于测试在发现缺陷方面的有效性
 - 但代码审查需要更多人力投入，因此效率较低
- ◆ 近年来，开发者开始关注技术债务（非缺陷问题），而测试对此类问题收效甚微。
 - 例如：可读性、可复用性、可维护性等
- ◆ 远程协作因"居家办公"需求兴起，使视障开发者能远程参与代码审查。后续幻灯片将展示若干应对策略。

例如：代码走查[22]

Table 3: Use of audio cues to convey awareness indicators in Follow mode (unless specified otherwise in the row)

Awareness Information	Non-Speech Indicator	Non-Speech Indicator Frequency	Speech Indicator	Speech Indicator Frequency	Built-in Visual Indicator
Viewport scrolls	Click wheel sound	Every scroll event	None	None	Screen scrolls
Scroll direction	Falling or rising tone depending on direction	When scrolling stops	None	None	Can be inferred from the scrolling viewport
Current Viewport	None	None	"Lines X to Y on screen"	When scrolling stops	Visible on screen
Cursor moves by single line to line N	Keyboard click	Every cursor move	"Line N"	1.5 seconds after cursor moves end	Cursor moves on screen
Cursor moves multiple lines to line N	Falling or rising tone depending on move direction	Every event	"Line N"	1.5 seconds after cursor moves end	Cursor moves on screen
Cursor moves by multiple lines to line N	Falling or rising tone depending on move direction	Every event	"Line N"	Every event	Cursor moves on screen
Selection	Depends on selection (key-board/mouse)	Every event	"Selection on line N"	1.5 seconds after selection is made	Selection visible on screen
Edits on follower's line	Keyboard type	For every character typed	None	None	Cursor moves on screen; edits visible on screen
Edits on follower's line (<i>Follow mode off</i>)	Keyboard type	For every character typed	"<collaborator> is editing the same line as you"	As long as edits continue on the same line	Cursor moves on screen; edits visible on screen
Edits within 5 lines of follower (<i>Follow mode off</i>)	Proximity sound	For every character typed	"<collaborator> is editing nearby"	As long as edits continue on the same line	Cursor moves on screen; edits visible on screen
Follow status	Pull and push sound	When follower starts and stops following leader	"You are now following <collaborator>"	Every event	None

```

16 ##comment
17 print('Hello')
18 print('World')
19 print('welcome')

```

BVI follower's cursor (grey) tethered to sighted leader's cursor (orange)

```

16 ##comment
17 print('Hello')
18 print('World')
19 print('welcome')

```

CLICK

```

16 ##comment
17 print('Hello')
18 print('World')
19 print('welcome')

```

CLICK CLICK 1.5s Delay Line 19

```

16 ##comment
17 print('Hello')
18 print('World')
19 print('welcome')

```

"Selection: print"

代码审查[6]

- ◆ 是一种高度结构化的正式审查流程
 - 企业中常称为 **静态测试**
 - 有效发现缺陷并识别进一步的流程改进机会
 - 日益失宠
- ◆ 核心要点
 - **会前**：作者向评审者讲解，评审者从测试角度阅读/质疑代码。
 - **会议中**：由代码作者带领审阅者逐行走代码，对照检查清单核对。解答疑问并消除歧义。每位审阅者在会前被分配预定义角色（如负责检查特定重点领域）。
 - 评审员在脑中“执行”代码
 - **会后**：作者跟进并修正代码。会议主持人核查所需整改措施是否落实到位。
- ◆ 特征
 - 正式、同步、代码检查为主、工具辅助为辅、权重较高⁹

代码审查

◆ 检查要点

- 算法效率

- 代码逻辑与正确性

- 例如：控制流与数据流、输入输出格式与顺序、参数与返回值类型及顺序等

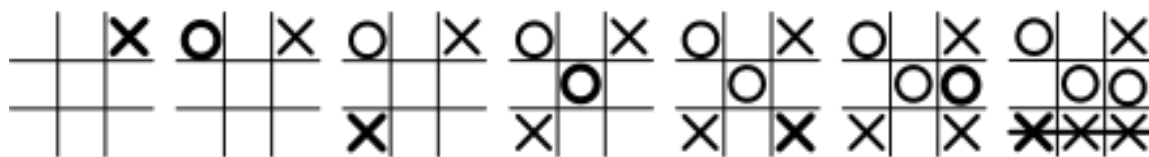
- 注释与代码一致性

◆ 制定自己的检查清单

MCR:

作者私人材料

井字棋（過三關）是一款双人游戏。游戏中存在一个3x3的网格。初始时网格为空。每个单元格可放置字符‘x’、字符‘o’或保持空置。当出现横向、纵向或对角线出现相同符号全数填满的情况，则该方获胜。例如图示中底部横排全被"x"符号占据，故"x"方获胜。



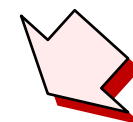
该程序允许两名玩家（玩家1和玩家2）轮流输入所选格子的坐标。每次接收输入后，程序需判断是否存在胜者，若有胜者则输出消息“胜！”。若填满全部9个格子后仍无胜者，则输出“平局！”。在显示“胜！”或“平！”后，立即显示 (l, sq) 整个井字棋棋盘并结束游戏。

MCR:

作者需提交给审稿人的材料

控制台界面：

输入内容以下划线标注



X = 玩家1 O = 玩家

2

玩家1：标记哪个单元格？ i:[0..2], j:[0..2]: 0 2

玩家2：标记哪个单元格？ i:[0..2], j:[0..2]: 0 0

玩家1：标记哪个单元格？ i:[0..2], j:[0..2]: 2 0

玩家2：标记哪个单元格？ i:[0..2], j:[0..2]: 1 1

玩家1：标记哪个单元格？ i:[0..2], j:[0..2]: 2 2

玩家2：标记哪个单元格？ i:[0..2], j:[0..2]: 1 2

玩家1：标记哪个单元格？ i:[0..2], j:[0..2]: 2 1 获胜！

O O

o o

X X X

任何意见/建议 (5分钟)

```
def is_win(game):
    win = False
    # Check rows
    if game[0][0] == game[0][1] == game[0][2] and (game[0][0] == 'X' or game[0][0] == 'O'):
        win = True
    if game[1][0] == game[1][1] == game[1][2] and (game[1][0] == 'X' or game[1][0] == 'O'):
        win = True
    if game[2][0] == game[2][1] == game[2][2] and (game[2][0] == 'X' or game[2][0] == 'O'):
        win = True
    # Check columns
    if game[0][0] == game[1][0] == game[2][0] and (game[0][0] == 'X' or game[0][0] == 'O'):
        win = True
    if game[0][1] == game[1][1] == game[2][1] and (game[0][1] == 'X' or game[0][1] == 'O'):
        win = True
    if game[0][2] == game[1][2] == game[2][2] and (game[0][2] == 'X' or game[0][2] == 'O'):
        win = True
    # Check diagonals
    if game[0][0] == game[1][1] == game[2][2] and (game[0][0] == 'X' or game[0][0] == 'O'):
        win = True
    if game[0][2] == game[1][1] == game[2][0] and (game[0][2] == 'X' or game[0][2] == 'O'):
        win = True
    return win
```

```
def main():
    game = [[' ' for _ in range(3)] for _ in range(3)] # Tic-tac-toe board
    player1 = 'X'
    player2 = 'O'
    turn = False # False for player 1's turn, True for player 2's turn. Player 1 first.
    print("X = Player 1")
    print("O = Player 2")
    for n in range(9):
        turn = not turn # Switch turns
        if not turn:
            print("Player 1: ", end="")
        else:
            print("Player 2: ", end="")
        print("Which cell to mark? i:[1..3], j:[1..3]: ")
        i, j = map(int, input().split())
        i -= 1
        j -= 1
        if not turn:
            game[i][j] = 'X'
        else:
            game[i][j] = 'O'
        if is_win(game):
            print("Win!")
            break # Terminate the game
    if n == 8: # All cells have been filled
        print("Tie!")
    # Show the game board
    for row in game:
        print(" ".join(row))

if __name__ == "__main__":
    main()
```

现代代码审查（MCR）

- ◆ MCR是一种轻量级且基于工具的*代码变更*审查（而非整段代码！），
 - 这是许多软件开发项目的标准做法
- ◆ MCR的特征
 - 非正式；工具驱动的流程；异步进行；专注于代码变更审查；比代码检查更轻量级
- ◆ 核心理念
 - 作者A对代码块C进行补丁P修改以解决问题，并通过电子渠道（如邮件、WhatsApp、微信或GitHub工具）将补丁P发送给评审组R
 - 每位审阅者R评估P：认可P或拒绝P
 - 作者A提交P；提交后其他R可进行审阅。
- ◆ 作者与审稿人交流观点、发现错误并探讨替代方案

为提交的代码变更提供更优的结构设计方案。

MCR流程

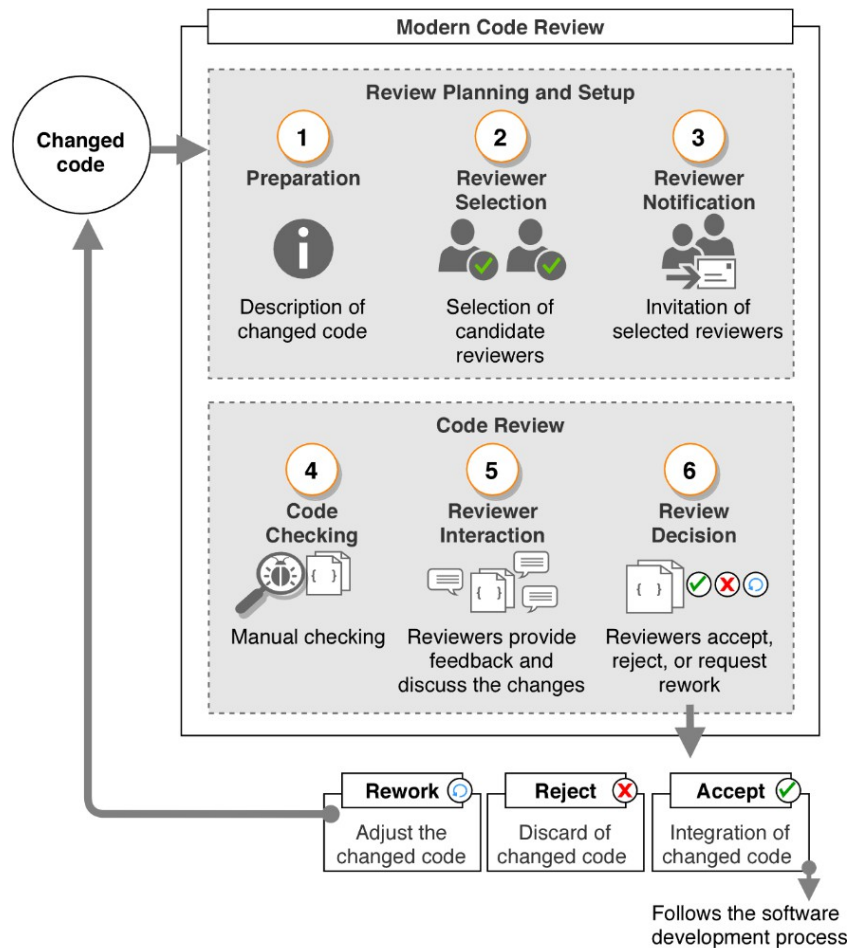


Fig. 1. An overview of the tasks of modern code review. [19]

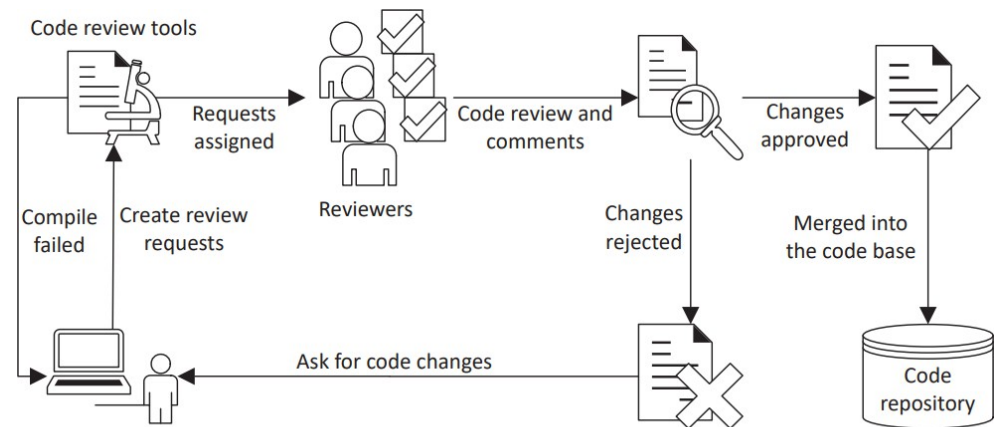


Fig. 1. An overview of the code review process. [20]

示例 [12]

Change 107871 - Merged

1

Implement EDP for a Spark standalone cluster

This change adds an EDP engine for a Spark standalone cluster. The engine uses the spark-submit script and various linux commands via ssh to run, monitor, and terminate Spark jobs.

Currently, the Spark engine can launch "Java" job types (this is the same type used to submit Oozie Java action on Hadoop clusters)

A directory is created for each Spark job on the master node which contains jar files, the script used to launch the job, the job's stderr and stdout, and a result file containing the exit status of spark-submit. The directory is named after the Sahara job and the job execution id so it is easy to locate. Preserving these files is a big help in debugging jobs.

A few general improvements are included:

- * engine.cancel_job() may return updated job status
- * engine.run_job() may return job status and fields for job_execution.extra in addition to job id

Still to do:

- * create a proper Spark job type (new CR)
- * make the job dir location on the master node configurable (new CR)
- * add something to clean up job directories on the master node (new CR)
- * allows users to pass some general options to spark-submit itself (new CR)

Partial implements: blueprint edp-spark-standalone

Change-Id: I2c84e9cdb75e846754896d7c435e94bc6cc397ff

Author	Alice <alice@redhat.com>
Committer	Alice <alice@redhat.com>
Commit	5698799ee3642a28797c6022dd35f228616764e1
Parent(s)	e23efe5471ed3e3ef3356918f80d91838f1c6585
Change-id	I2c84e9cdb75e846754896d7c435e94bc6cc397ff

2

History

Alice Uploaded patch set 1

.....

Bob

Patch Set 1:

sahara/service/edp/job_manager.py

Line 68:

should this be guarded with:
if job_info.get('status') in job_utils.terminated_job_states:
just in case 'status' doesn't exist?

Alice

Patch Set 4:

The patch LGTM, apart from the small comment on the commit message. One important question, though, is about the data sources. How is input and output specified for each job submitted through Spark EDP? Spark does not support Swift for now, so I would expect only HDFS to be available.

Owner	Trevor McKay			
	Bob	Alice	John	Rob
Reviewers	Edward	Sam	Ryan	Alex
	Enzo	Frank		
Project	5698799ee3642a28797c6022dd35f228616764e1			
Branch	e23efe5471ed3e3ef3356918f80d91838f1c6585			

3

Files	Comments	
sahara/plugins/spark/plugin.py	33	
sahara/service/edp/job_utils.py	46	
sahara/service/edp/oozie/engine.py	46	
sahara/service/edp/job_utils.py	18	
sahara/service/edp/oozie/oozie.py	7	
sahara/service/edp/resources/launch_command.py	66	
sahara/service/edp/spark/engine.py	161	
sahara/tests/unit/service/edp/spark/__init__.py	0	
sahara/tests/unit/service/edp/spark/test_spark.py	383	
sahara/tests/unit/service/edp/test_job_manager.py	10	

4

请看下一张幻灯片

5

示例[12]

在变更中展开文件

```
class FormPostTest(ObjectStorageFixture):

    @classmethod
    def setUpClass(cls):
        super(FormPostTest, cls).setUpClass()
        cls.key_cache_time = (
            cls.objectstorage_api_config.tempurl_key_cache_time)
        cls.tempurl_key = cls.behaviors.VALID_TEMPURL_KEY
        cls.object_name = cls.behaviors.VALID_OBJECT_NAME
        cls.object_data = cls.behaviors.VALID_OBJECT_DATA
        cls.content_length = str(len(cls.behaviors.VALID_OBJECT_DATA))
        cls.http_client = HTTPClient()
        cls.redirect_url = "http://example.com/form_post_test"
```

```
class FormPostTest(ObjectStorageFixture):

    @classmethod
    def setUpClass(cls):
        super(FormPostTest, cls).setUpClass()
        cls.key_cache_time = (
            cls.objectstorage_api_config.tempurl_key_cache_time)

        cls.object_name = cls.behaviors.VALID_OBJECT_NAME
        cls.object_data = cls.behaviors.VALID_OBJECT_DATA
        cls.content_length = str(len(cls.behaviors.VALID_OBJECT_DATA))
        cls.http_client = HTTPClient()
        cls.redirect_url = "http://example.com/form_post_test"
```

Alice

Apr 22, 2015

Should there be a default value for the cls.tempurl_key, or should the fixture assert if keys_set is empty? All of the tests depend on the attribute, but for whatever reason, if the X-Account-Meta-Temp-Uri-Key is not present in the headers, the attribute will not exist, and the tests will error out in an ungraceful manner.

[Reply](#) [Quote](#) [Done](#)

Bob

Apr 22, 2015

So the check_account_tempurl_keys method will first check to see if the account keys are set and if they aren't, it will set them to some defaults. If, for some reason, it fails to set them properly, keys_set should be False instead of True. So, what I will do is have an else statement here which will raise an Exception but in all likelihood it would fail in the behaviors beforehand.

[Reply](#) [Quote](#) [Done](#)

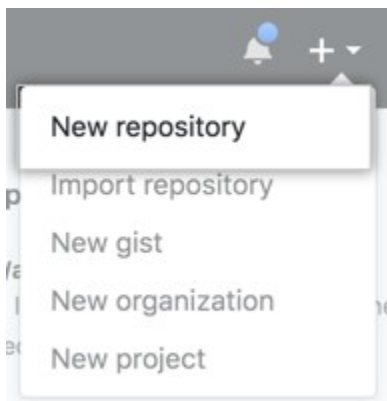
```
40 @ObjectStorageFixture.required_features('formpost')
41 def test_object_formpost_redirect(self):
42     """
```

```
44 @ObjectStorageFixture.required_features('formpost')
45 def test_object_formpost_redirect(self):
46     """
47
```

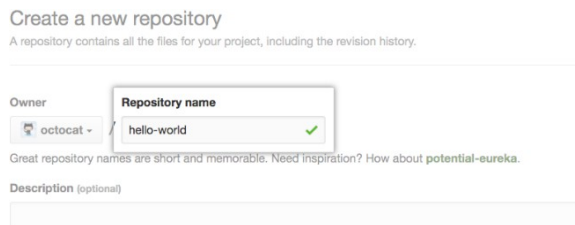
示例 [17]



1. 在任意页面的右上角，使用下拉菜单，选择**新建存储库**。



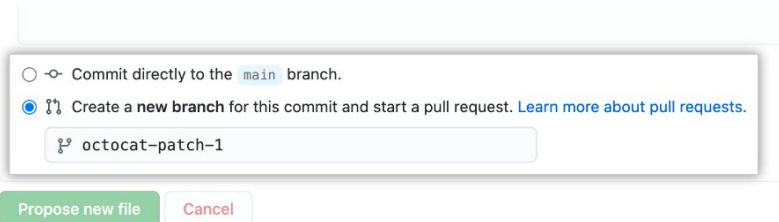
2. 为您的存储库输入简短易记的名称。例如"hello-world"。



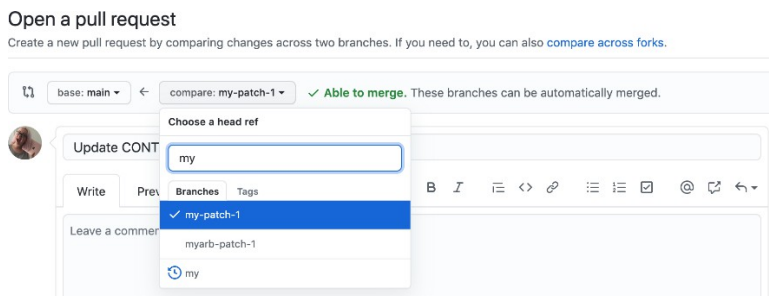
3. 对文件进行所需修改。



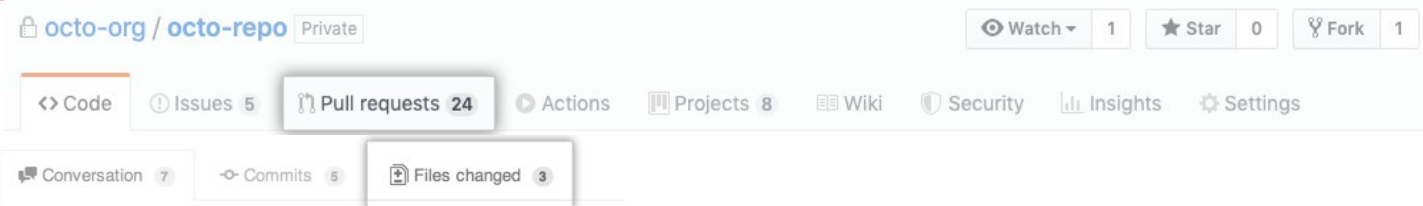
4. 决定是否将您的提交添加到当前分支或新建分支。



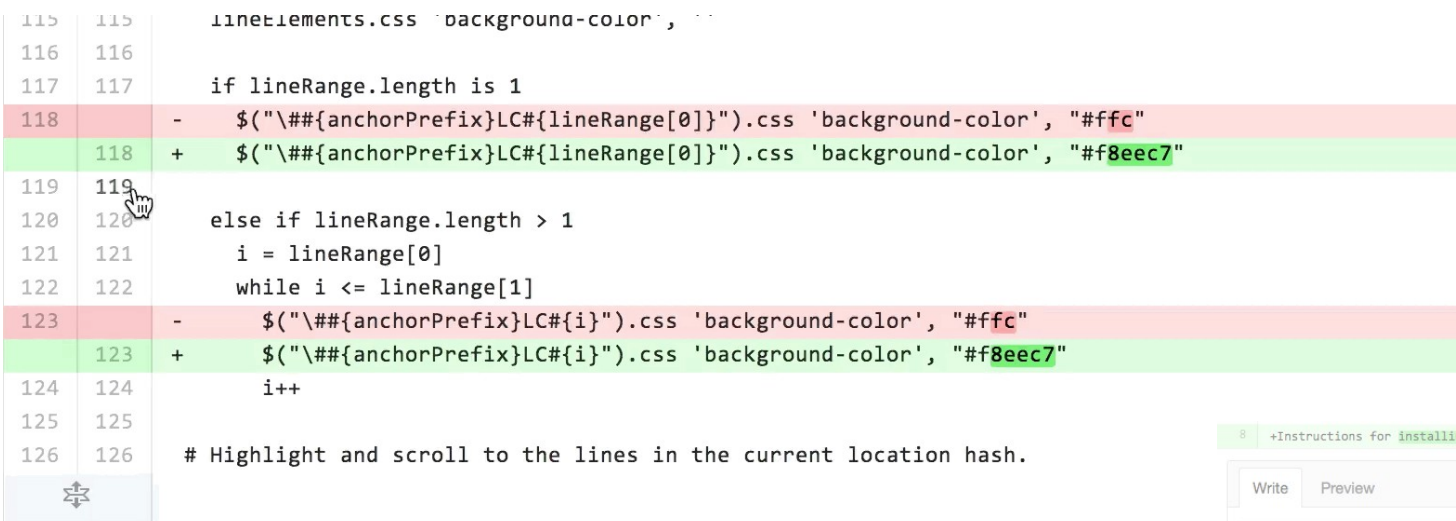
5. 创建拉取请求以提出并协作修改仓库内容。



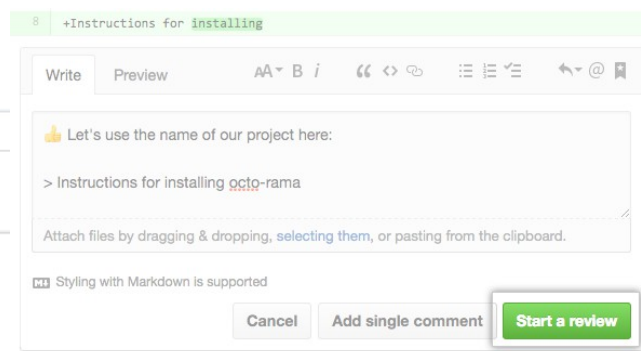
示例 [17]



6. 将鼠标悬停在需要添加注释的代码行上，点击蓝色注释图标。若需添加多行注释，请点击并拖动选择行范围，再点击蓝色注释图标。



7. 完成后点击"开始审查"。若已启动审查，可点击"添加审查评论"。



现代代码审查的特征

◆ 非正式

- 由临时组建的评审小组进行，成员无预设角色，评审流程不遵循正式程序

◆ 工具驱动的后勤支持

- 电子邮件、WhatsApp、微信广受欢迎。
- 开源工具：GitHub、Gerrit（用于Git）、ReviewBoard、Phabricator

◆ 异步协作

- 作者与审阅者无需同时处理同一任务（但不可过度延迟）

◆ 审阅聚焦于补丁（代码变更，增量部分）

评审/评论/帖子的目的 [1]

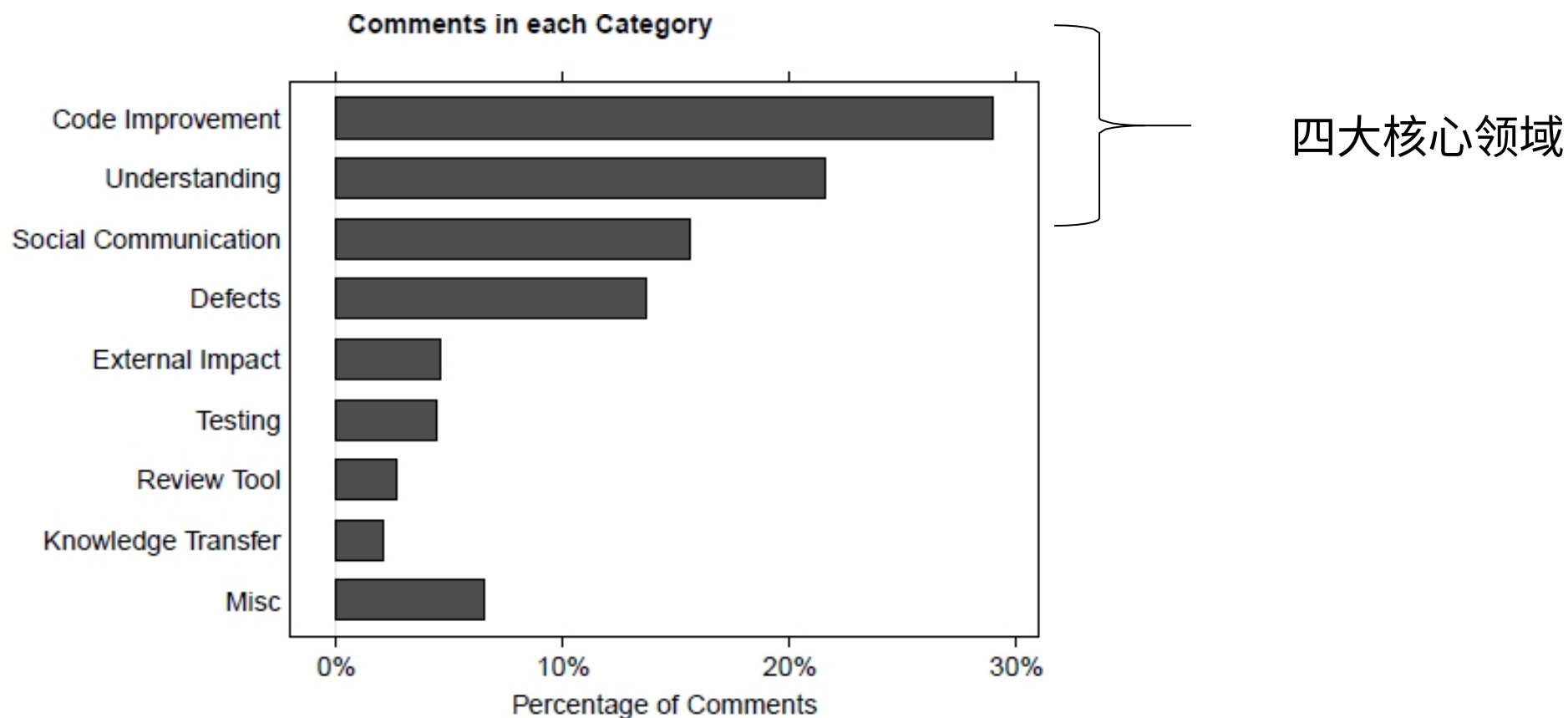


Fig. 4. Proportion of comments by card sort category.

理解代码的目的是什么？ [1]

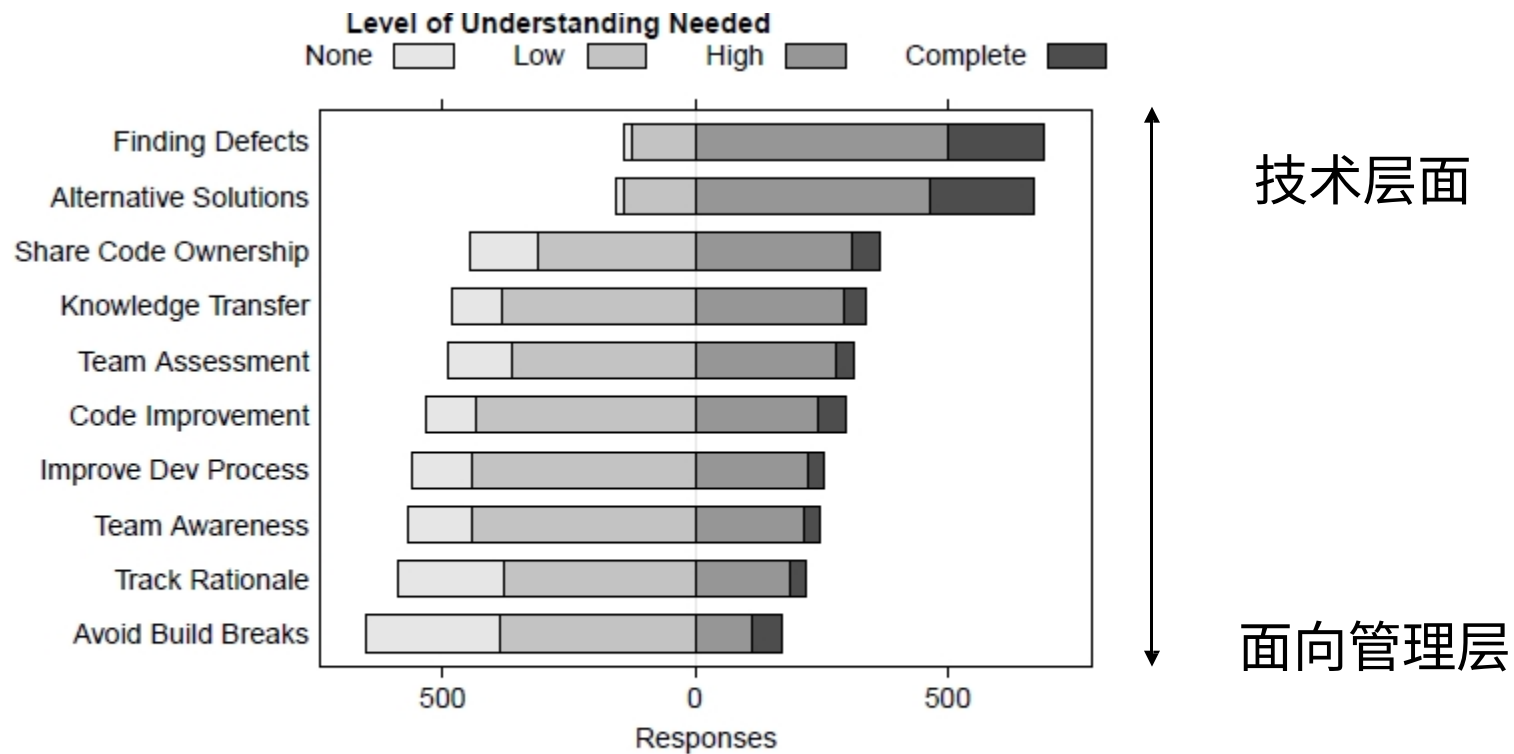


Fig. 5. Developers responses in surveys of the amount of code understanding for code review outcomes.

应对困惑的策略 [14]

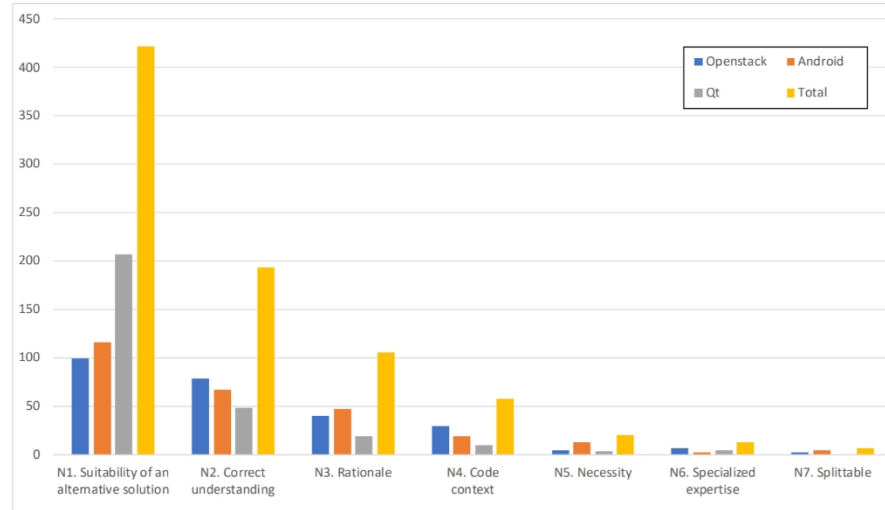
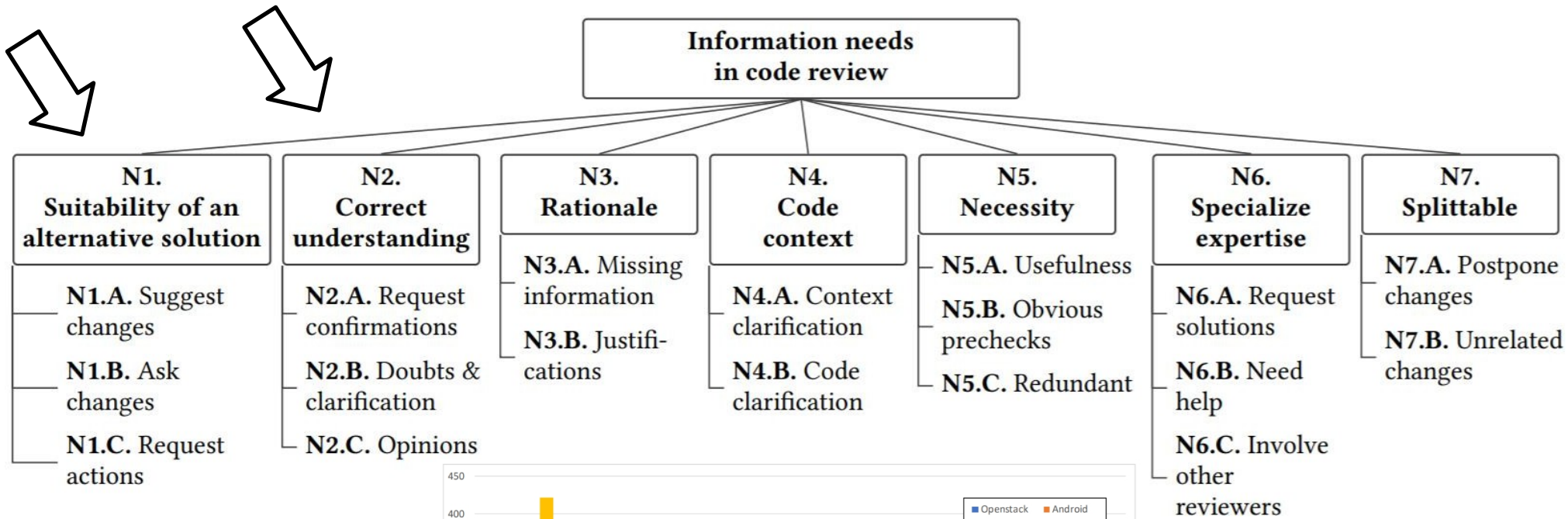
THE REASONS, IMPACTS AND COPING STRATEGIES DEVELOPERS USE TO DEAL WITH CONFUSION; IN THE PARENTHESIS ARE THE NUMBERS OF CARDS.

	Review process <i>18 topics (120)</i>	Artifact <i>15 topics (300)</i>	Developer <i>15 topics (124)</i>	Link <i>9 topics (177)</i>
Reasons <i>30 topics (507)</i>	Organization of work (17) Issue tracker, version control (7) Unnecessary change (6) Not enough time (3) Dependency between changes (3) Code ownership (2) Community norms (2)	Missing rationale (66) Discussion of the solution: non-functional (49) Unsure about system behavior (37) Lack of documentation (29) Discussion of the solution: strategy (29) Long, complex change (25) Lack of context (19) Discussion of the solution: correctness (14) Impact of change (11) Irreproducible bug (6) Lack of tests (5)	Disagreement (18) Communicative intention (9) Language issues (3) Propagation of confusion (3) Fatigue (1) Noisy work environment (1)	Lack of familiarity with existing code (47) Lack of programming skills (40) Lack of understanding of the problem (21) Lack of understanding of the change (17) Lack of familiarity with the technology (14) Lack of knowledge about the process (3)
Impacts <i>14 topics (98)</i>	Delaying (31) Decreased review quality (11) Additional discussions (11) Blind approval (8) Review rejection (4) Increased development effort (4) Assignment to other reviewers (2)	Better solution (1) Incorrect solution (1)	Decreased confidence (10) Abandonment (6) Frustration (5) Anger (2) Propagation of confusion (2)	
Coping strategies <i>13 topics (116)</i>	Improved organization of work (5) Delaying (2) Assignment to other reviewers (1) Blind approval (1)	Small, clear changes (4) Improved documentation (4)	Information requests (36) Off-line discussions (12) Providing/accepting suggestions (10) Disagreement resolution (6)	Improved familiarity with existing code (28) Testing the change (5) Improved familiarity with the technology (2)



没有特别有效的策略

评审者需要哪些信息? [12]



代码重构的审查补丁如何？

- ◆ 施乐公司的工业案例研究[23]
- ◆ "应对糟糕的设计和编码风格是开发者在代码变更中进行重构的主要驱动力。然而，功能变更和修复缺陷的活动也常常促使开发者重构代码。"

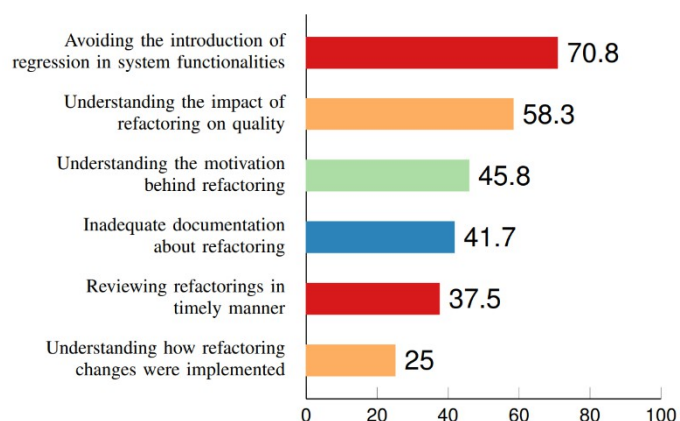


Figure (4) Challenges faced by developers when reviewing refactoring.

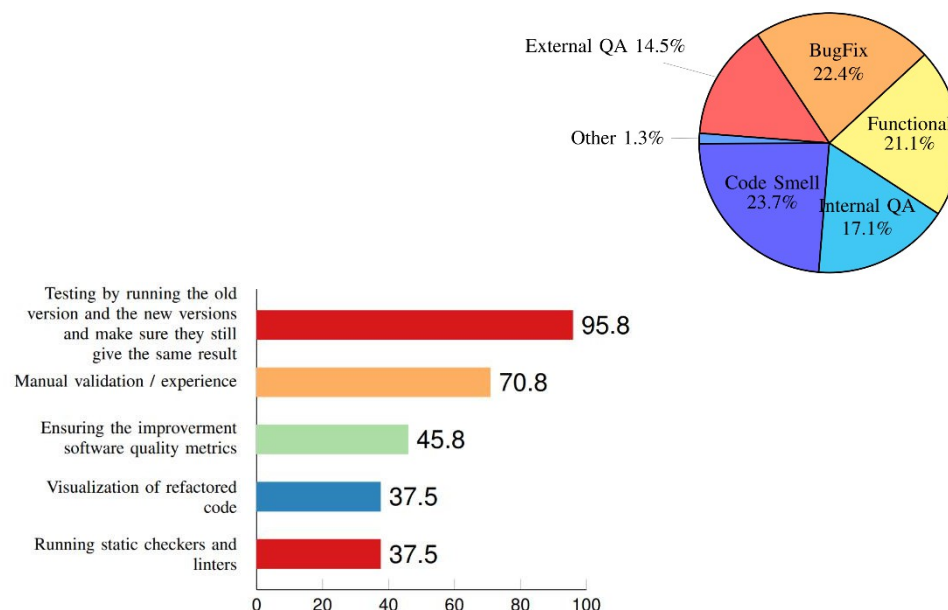


Figure (5) Mechanisms used to ensure the correctness after the application of refactoring.

知识转移呢？

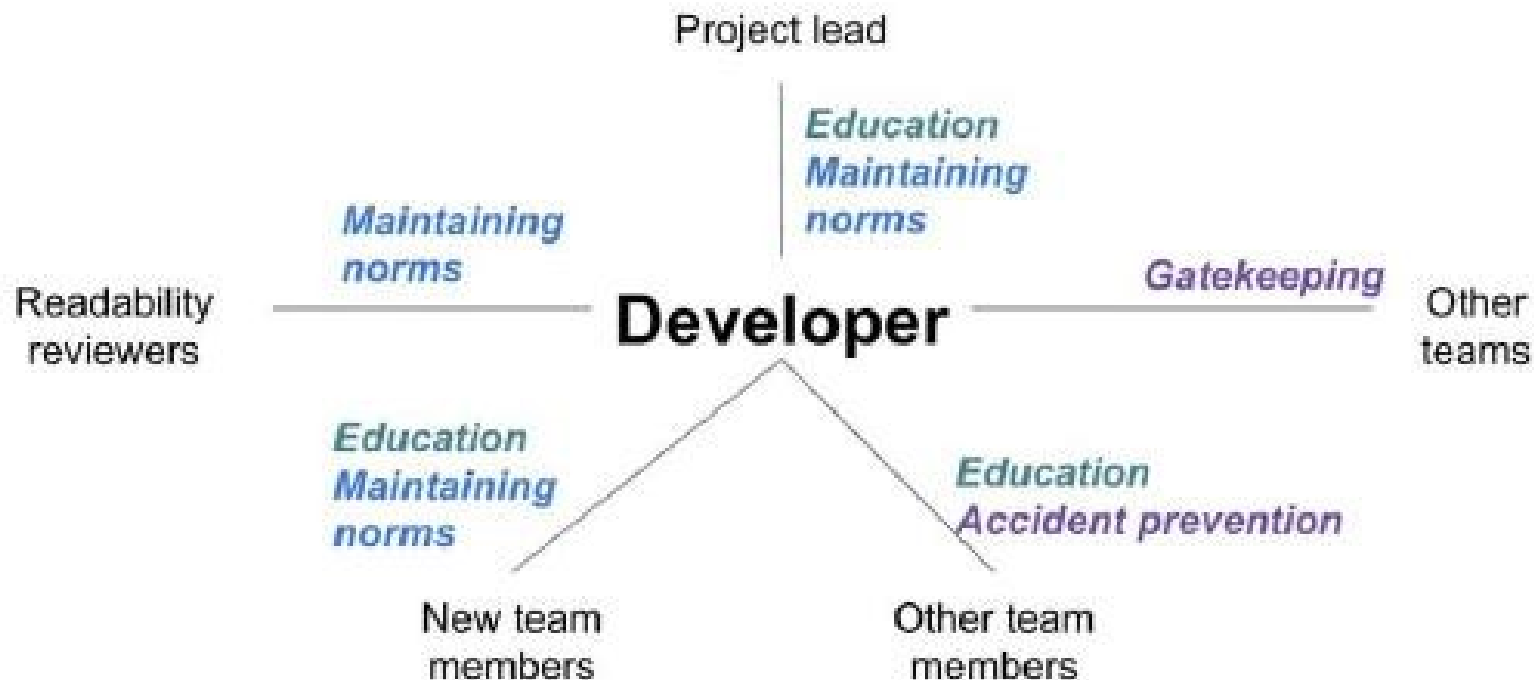
- ◆ 增加代码审查轮次能否提升开发者的代码质量？
 - 遗憾的是，目前无论是单个项目内部还是跨项目层面，均未发现显著证据。[16]
- ◆ 那么，让我们看看大型科技公司的代码审查机制。请参阅后续几张幻灯片。

代码审查最佳实践[1, 3, 7]

- ◆ 当代同行评审遵循轻量化、灵活的流程
- ◆ 代码审查应在早期（提交变更前）、及时且频繁地进行
- ◆ 审阅者事先了解上下文和代码，能更快完成审阅并为作者提供更有价值的反馈
- ◆ 变更规模较小
- ◆ 审阅者数量较少（1至5人）
- ◆ 阅读代码及补丁的非正式讨论与评论
- ◆ 审查不仅限于发现组级缺陷：还包括问题解决（方案开发）、代码可读性与可维护性、遵循规范（所属公司标准）、事故预防、把关（召回率）

代码审查中主持人的角色？

MCR最佳实践[3]



- ◆ 上图展示了谷歌代码项目中开发者（即作者）与审阅者之间的关系

谷歌MCR最佳实践[3]

- ◆ 建议评审人：请能够担任评审的人员轮流担任评审，采用循环轮换的方式。
 - 避免长期过度占用团队中最佳人选的精力
- ◆ 通过代码分析补充人工评审
 - 应用风格检查器、代码检查工具
 - 例如：Java的Checkstyle
- ◆ 评审频率：每周4小时评审（及4次评审）（中位数）
- ◆ 代码审查速度：作者等待1-5小时
- ◆ 变更代码规模：1行代码（约10%）、单个文件（约30%）、

少于10个文件（约90%）。

MCR最佳实践

◆ 候选审阅者？

- 代码所有者
- 曾对代码进行过修改的开发者
- 先前参与代码开发的开发者（例如通过结对编程产出代码者）
- 经验丰富的审阅者
- 在其他代码中开发过类似变更片段功能的开发者
- ...

重大变更请求（MCR）问题

◆ 代码质量与代码规模

- 复杂度较高的代码在代码审查中获得的讨论和反馈较少。
- 代码评审反馈越少，发布后出现的缺陷就越多

◆ 距离

- 导致代码审查流程延迟或引发误解
- 方面1：地理因素
 - 作者与审阅者之间存在较大物理距离
- 方面二：组织问题
 - 审稿人来自不同团队或承担不同职责

◆ 社交互动

- 语气：负面语气评论的实用性较低
- 权力：要求他人修改评论。作者感到不满

紧急修复 vs. 全面解决方案 vs. 高级草图)

◆ 背景

- 因代码变更预期不匹配导致的误解（例如：可有可无的功能）

无效代码变更请求的其他问题：MCR反模式[18]

TABLE I: MCR anti-patterns and their associated symptoms and consequences.

MCR anti-pattern	Symptoms	Potential Consequences
Confused reviews (CR)	- Reviewers ask question(s) about the rationale or the solution approach of the patch. - Reviewers express incertitude in review comments.	<i>Process</i>: Confusion decreases the efficiency and the effectiveness of the review [6], [10]. <i>Artifact</i>: The patch may still have poor quality after the review as reviewers may not fully understand the patch [6]. <i>People</i>: Reviewers may feel frustrated and may express negative sentiments due to the confusion [6].
Divergent reviews (DR)	- The review decisions do not reach consensus. - The review scores are diverged. - Reviewers post conflict review comments to each others.	<i>Process</i>: The divergence can slow down the integration process [16] <i>Artifact</i>: It is correlated with negative development outcomes (i.e., patches without changing) [16] <i>People</i>: The divergence increases contributors' likelihood of leaving the communities and lead to the poor team performance [11], [29]
Low review participation (LRP)	- The patch does not have other developers as a reviewer. - The patch receive few/short comments. - The patch does not receive prompt/timely feedback from reviewers.	<i>Process</i>: The lack of review participation has a negative impact on review efficiency and effectiveness [30]. <i>Artifact</i>: Patches with low number of reviewers can be more defect-prone [7] <i>People</i>: The inefficient reviewer feedback could make the patch author forget the change. [5]

TABLE II: The detection results of MCR anti-patterns.

Anti-pattern	# instances	Kappa	Agreement
Confused Reviews (CR)	21 	0.93	Perfect
Divergent Reviews (DR)	20 	1.0	Perfect
Low Review Participation (LRP)	32 	1.0	Perfect

TABLE III: The prevalence of MCR anti-patterns.

Category	Count
Code reviews affected by one anti-pattern	67 
Code reviews affected by two anti-patterns	21 
Code reviews affected by three anti-patterns	4 

是否也应审查测试代码？

◆ 简单回答是"应该"，但需调整关注重点
且频率较低（需与生产代码同步审查）

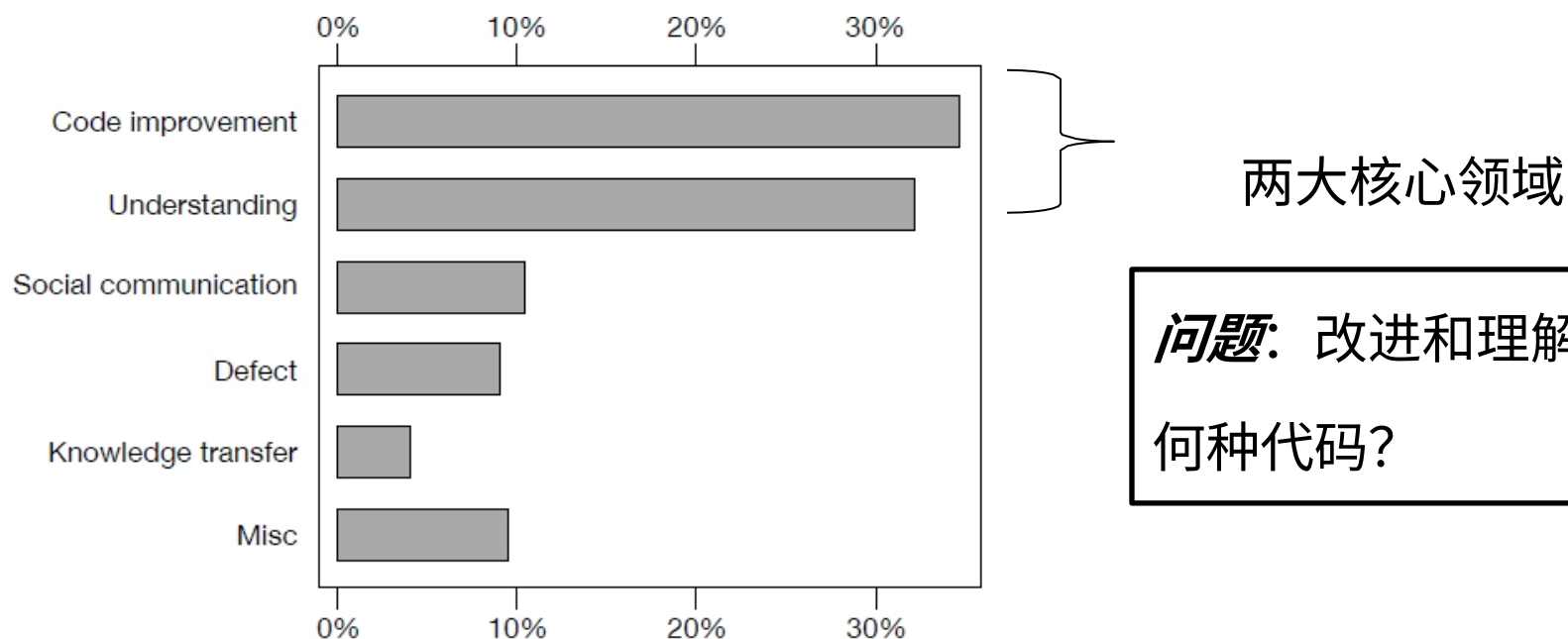


Figure 1: The outcomes of comments in code review of test files, after manual analysis in 600 comments (CL=99%, CI=5). [4]

通过审查测试代码来理解和改进哪些类型的代码? [15]

由德国研究人员对92名开发者开展。

INTERVIEWEES' EXPERIENCE (IN YEARS) AND WORKING CONTEXT

ID	Developer	Reviewer	Working context	Applying TDR
P1	8	8	OSS	Almost never
P2	3	3	Company A	Almost Always
P3	15	15	Company A	Almost Always
P4	10	10	Company B	Almost Never
P5	10	5	Company C	Always
P6	3	2	Company D	Sometimes
P7	16	16	Company E	Always
P8	4	4	Company F / OSS	Sometimes
P9	3	3	Company G / OSS	Never

Finding 1. *In reviews in which tests are presented first, participants found significantly more bugs in test code, yet fewer maintainability issues in production code. The production bugs and test maintainability issues found is stable across the treatments.*

Finding 3. *Perceived problems with TDR: Developers report to (1) consider tests as less important than production, (2) not being able to extract enough knowledge from tests, (3) not being able to start a review from tests of poor quality, and (4) being comfortably used to read the patch as presented by their code review tool.*

Finding 4. *Perceived advantages of TDR: Developers report that TDR (1) allows them to have a concise, high-level overview of the code under test and (2) helps them in being more testing-oriented, hence improving the overall test code quality.*

现代代码审查：

课堂活动 MCR-2

- ◆ 组成4-5人小组
 - 代码作者A：1名学生
 - 代码审阅者R：无预设角色
 - 成员间协商确定评审工具（我们使用Github支持）
- ◆ 时间线（单位：分钟）
 - 00-15：A 在 Github 平台为代码 C 准备补丁 P，附上补丁说明，并通过电子渠道通知组员
 - 15-25分钟：R审阅P并在Github发布评论。审阅完成后通过电子渠道通知
 - 15-25：A 逐条回应 R 提出的疑问并修订补丁 P' 进行迭代修订
 - 25-30: R 批准 P，A 应用最终补丁 P' 获得最终修正代码 C2
 - 30: A将修正后的C2附带评论（图片或Github仓库链接？）提交至Canvas讨论区供同行共享，同时将C2作为提交

自学家庭练习 [无需提交]

◆ 阅读参考文献[1]

- 探索提升团队协作意识的方法
- 第5页资深开发者总结的代码审查五大益处是什么？
- 第5-6页中，团队可采取哪些措施促进知识转移？
- 第7页中，如果代码审查质量较低，会出现哪些问题？
- 第9页提出了哪些建议？你是否认同？

◆ 阅读参考文献[5]

- 第23页，幻灯片演示作者认为该代码审查支持工具具备哪些核心功能？
幻灯片演示作者认为该工具具备？

◆ 自我反思

- 若公司要求你在项目中实施MCR但未提供额外预算（\$），你将如何实施MCR？
- 距离是MCR团队面临的主要问题。当团队成员分布在不同城市且可能拥有不同文化背景）如何开展MCR？

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